

CHAPTER 1

GROUP ACTIVITY Solving a Cubic Equation

If a , b , and c are real numbers with $a \neq 0$, then the quadratic equation $ax^2 + bx + c = 0$ can be solved by a variety of methods, including the quadratic formula. How can we solve the cubic equation

$$ax^3 + bx^2 + cx + d = 0, \quad a \neq 0 \quad (1)$$

and is there a formula for the roots of this equation?

The first published solution of equation (1) is generally attributed to the Italian mathematician Girolamo Cardano (1501–1576) in 1545. His work led to a complicated formula for the roots of equation (1) that involves topics that are discussed later in this text. For now, we will use Cardano's method to find a real solution in special cases of equation (1). Note that because a is nonzero, we can always multiply both sides of (1) by $1/a$ to make the coefficient of x^3 equal to 1.

CARDANO'S METHOD FOR SOLVING A CUBIC EQUATION

Let $x^3 + bx^2 + cx + d = 0$

Example problem: $x^3 - 6x^2 + 6x - 5 = 0$. Steps will be in red.

Step 1. Substitute $x = y - b/3$ to obtain the *reduced cubic* $y^3 + my = n$.

$x = y - \frac{-6}{3}$ or $x = y + 2$. The equation becomes

$$(y + 2)^3 - 6(y + 2) + 6(y + 2) - 5 = 0,$$

which simplifies to $y^3 - 6y = 9$; $m = -6$, $n = 9$.

Step 2. Define u and v by $m = 3uv$ and $n = u^3 - v^3$. Use $v = \frac{m}{3u}$ to write

$$n = u^3 - \left(\frac{m}{3u}\right)^3$$

Multiply both sides by u^3 to obtain an equation quadratic in u^3 . Solve for u^3 by factoring or by using the quadratic formula. Then solve for u , and find the associated value of v .

$v = \frac{-6}{3u}$ or $v = -\frac{2}{u}$; $9 = u^3 - \left(-\frac{2}{u}\right)^3 = u^3 + \frac{8}{u^3}$. Multiply both sides by u^3 to obtain $u^6 - 9u^3 + 8 = 0$; solve by factoring to get $u = 2$ (in which case $v = -1$) or $u = 1$ (in which case $v = -2$).

Step 3. Using either of the solutions found in step 2,

$$x = y - \frac{b}{3} = u - v - \frac{b}{3}$$

is a solution to $x^3 + bx^2 + cx + d = 0$

For $u = 2$, $v = -1$, $x = 2 - (-1) - \frac{-6}{3} = 5$ (Solution)

(A) The key to Cardano's method is to recognize that if u and v are defined as in step 2, then $y = u - v$ is a solution of the reduced cubic. Verify this by substituting $y = u - v$, $m = 3uv$, and $n = u^3 - v^3$ in $y^3 + my = n$ and show that the result is an identity.

(B) Use Cardano's method to solve

$$x^3 - 6x^2 - 3x - 8 = 0 \quad (2)$$

Use a calculator to find a decimal approximation of your solution and check your answer by substituting this approximate value in equation (2).

(C) Use Cardano's method to solve

$$x^3 - 6x^2 + 9x - 6 = 0 \quad (3)$$

Use a calculator to find a decimal approximation of your solution and check your answer by substituting this approximate value in equation (3).

(D) In step 2 of Cardano's method, show that u^3 is real if $\left(\frac{n}{2}\right)^2 \geq \left(\frac{-m}{3}\right)^3$.